

Life in polar regions

Although they live at opposite ends of the earth, the animals that inhabit the Arctic and Antarctic share many adaptations. Resilience to extreme cold is first and foremost among these, but almost as important is the ability to cope with a highly seasonal food supply. For some animals, winter is a good time for catching food but, for most, hunger and cold make the long winter months a critical time of year. Such testing conditions mean that, in comparison with other parts of the world, the poles are inhabited by very few animal species. However, those that do thrive can be extraordinarily numerous.

Coping with cold

Warm-blooded animals have to maintain a constant body temperature, which means that combating heat loss is a major priority in the polar environment. Cold-blooded animals can function with a fluctuating body temperature, but even they have limits—in subzero conditions, they can freeze solid. Fish are particularly at risk of freezing, for while their body fluids normally freeze at about 30.6°F (-0.8°C) polar sea water is often slightly colder still. To help prevent freezing, the blood of many cold-blooded species contains proteins that lower its normal freezing point. Some insects can survive at -49°F (-45°C) without any ice forming in their bodies.

Since mammals and birds cannot afford to let their internal temperature fall even slightly, they need insulation to keep warm. Fur and feathers are among the finest insulating



THE ANTARCTIC'S ICE FISH

In the Antarctic, fish live in a world that is often roofed by ice. The high level of oxygen in cold water allows some Antarctic fish to survive without any hemoglobin in their blood. As a result, their blood is almost colorless rather than red.

materials that nature has devised, but many polar animals, such as whales, seals, and penguins, have additional insulation in the form of blubber—a layer of yellowish fat that is laid down under the skin. Blubber, which can be up to 12 in (30 cm) thick, is such an effective insulator that all these animals feel cold to the touch even when their internal body temperature is above 100.5°F (38°C). Blubber is particularly useful in the sea because water conducts heat away from the body 25 times faster than air. It also has another valuable function: because fat contains lots of energy, blubber can be used as a food reserve when supplies are low.



SLEEPING ON ICE

A polar bear cub rests at its mother's side. Polar bears have long body fur, and their feet have furry soles. This keeps them warm on the ice, while a thick layer of blubber enables them to retain body heat while swimming in the ocean.

Surviving under ice

Whales and seals face problems during the long polar winter because sea ice restricts their access to air. They dive under the ice to feed, but they must then surface to breathe. Some species avoid the problem by moving to lower latitudes. Those that remain behind survive either by maintaining breathing holes or by congregating in polynyas—areas where the wind and currents keep the water ice free.

Seals start making breathing holes when the ice is thin, rasping away at it with their teeth. As the depth of the ice increases with the progress of winter, they continue to visit and work on their holes to ensure they remain clear. The Weddell seal, which lives farther south than any other species, spends so much time keeping its breathing holes open that its teeth develop distinctive patterns of wear, and by late winter, its breathing holes can be 6½ ft (2 m) deep. The seals have to find their holes in almost complete darkness because during the Antarctic winter the sun stays below the horizon for weeks.

PAUSE FOR BREATH

A Weddell seal heads up toward a breathing hole in the ice. Weddell seals can stay underwater for over an hour, making return trips of up to 6 miles (10 km) before they have to come up for air.

Whales rarely make breathing holes. Instead, they head for polynyas, where they can come up for air in open water. This less laborious strategy means they are not tied to one place, but it does have its dangers: groups of whales can become ensnared in shrinking polynyas, unable to reach the next stretch of open water. There are records of narwhals—the world's most northerly whales—being trapped in their hundreds, making them easy targets for hunters.

Changing color

In the treeless Arctic tundra, camouflage is one of the most effective ways both of avoiding attack and of making an attack unseen. The summer and winter landscapes look so different that many tundra animals change their camouflage twice a year. The Arctic fox is a classic example: its summer coat is usually brownish gray, but in early fall it turns white; in spring, the process is reversed so that the fox blends with the rapidly thawing tundra. In some parts of the far north—particularly western Alaska and northern Greenland—Arctic foxes develop a bluish winter coat instead. Some researchers have suggested that this blue coat is an adaptation to coastal landscapes, where there is less winter snow, but as these foxes have been widely introduced by fur farmers, the theory is difficult to prove.

The stoat, a member of the weasel family, changes coat color in a similar way, as do ptarmigans and many other tundra birds. Some, such as the snowy owl, keep their white plumage all year round, which suggests that good camouflage is most important in winter and less so in summer when food is easier to find.

